



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024
ME1101-1- ENGINEERING METROLOGY
(ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- What is the distance between supports for a simply supported beam? (where L is length of the bar)

A) 0.557 L	B) 0.554 L
C) 0.575 L	D) 0.577 L
- By which material Prototype metre (line standard) bar is made of?

A) Bronze	B) Platinum
C) Platinum-bronze alloy	D) Platinum-iridium alloy
- Which standard is fixed and used for industrial laboratories?

A) International standard	B) Primary standard
C) Secondary standard	D) Working standard
- Error of measurement =

A) True value – Measured value	B) Precision – True value
C) Measured value – Precision	D) None of the above
- Which of the following is true for uses of comparators?

A) Can't be used in mass production	B) Not suitable for inspection purposes
C) Can be used as working gauge	D) Slow rate of working
- What is dedendum for external threads?

A) Radial distance between pitch and minor cylinder	B) Radial distance between major and pitch cylinder
C) Radial distance between major and minor cylinder	D) Axial distance between major and pitch cylinder
- Which of the following is not true about effective diameter?

A) Also known as pitch diameter	B) It decides quality of fit between nut and screw
C) This is the diameter of minor cylinder	D) It is a very important dimension for screw threads
- Circle which corresponds to section of equivalents cylinder by a plane normal to wheel axis is known as

A) Pitch cylinders	B) Pitch diameter
C) Pitch circle	D) Pitch point
- The output of a transducer must be _____

A) Low	B) Medium
C) High	D) zero
- Electrical transducers are _____

A) small and non-portable	B) large and non-portable
C) small and compact	D) large and portable
- The power needs of electrical transducers is _____

A) Maximum	B) minimum
C) zero	D) infinite

12. A transducer must be _____
 A) quick in response B) slow in response
 C) medium in response D) very slow in response
13. Strain gauge measurement involves _____
 A) Wheatstone bridge B) Kelvin bridge
 C) De Sauty's bridge D) Anderson bridge
14. Strain gauge works on the principle of _____
 A) piezo-electric effect B) barkhausen criterion
 C) piezo- resistive effect D) feedback element effect
15. Pressure measurement devices make use of _____
 A) non-elastic member B) elastic member
 C) bendable member D) non-bendable member
16. Bourdon tubes are _____
 A) very highly sensitive to shock B) not sensitive to shock
 C) less sensitive to shock D) more sensitive to shock
17. The difference between the maximum material limits of the mating parts is called _____
 A) Deviation B) Allowance
 C) Tolerance D) Actual deviation
18. Minimum clearance is the difference between _____ size of the hole and the _____ size of the shaft.
 A) minimum, maximum B) minimum, minimum
 C) maximum, maximum D) maximum, minimum
19. Maximum clearance is the difference between _____ size of the hole and the _____ size of the shaft.
 A) minimum, maximum B) minimum, minimum
 C) maximum, maximum D) maximum, minimum
20. Maximum interference is the magnitude of the difference between _____ size of the hole and the _____ size of the shaft.
 A) minimum, maximum B) minimum, minimum
 C) maximum, maximum D) maximum, minimum

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain any one material standards of length.	4	L2	1	1
	b) Compare accuracy with precision.	4	L4	1	2
	c) Sketch and explain the terminology of gear.	8	L2	2	2
2.	a) Explain with a neat sketch micro-optic autocollimator.	8	L2	1	1
	b) Describe with a neat sketch Solex comparators.	8	L1	2	2
3.	a) Define the following terms; calibration, threshold, sensitivity and hysteresis.	8	L1	1	2
	b) Explain in detail the characteristics of comparators.	8	L2	2	1
Unit – II					
4.	a) Give the transducers classifications.	3	L1	3	2
	b) Difference between active & passive transducer	5	L3	3	1
	c) With sketch explain the working principle of hydraulic dynamometer.	8	L2	4	2
5.	a) Explain the working of LVDT and highlight its applications.	8	L2	3	2
	b) Obtain the step-by-step procedure of strain gauge preparation and mounting of strain gauges.	8	L2	4	1
6.	a) With a circuit diagram explain vacuum tube amplifiers	8	L2	3	1
	b) Sketch and explain Prony brake dynamometer	8	L2	4	1

Unit – III

7. a) Describe with necessary sketches hole basis system and shaft basis system. 6 L1 5 1
- b) Calculate the tolerances on the hole and the shaft for a precision running fit designated by $50H_7/g_6$. You may use the following aid for solving the problem.
- i) 50mm lies between the step size of 30-50mm.
 - ii) $i = 0.45(D)^{1/3} + 0.001D$
 - iii) Fundamental deviation for the 'H' hole = 0 and Fundamental deviation for g shaft = $-2.5D^{0.34}$
 - iv) IT 7 = 16i and IT 6 = 10i
- State the actual maximum and minimum sizes of the hole and shaft and maximum and minimum clearances. 10 L3 5 2
8. a) Obtain the Taylor's principles of gauge design. 6 L1 5 1
- b) Determine the tolerances on the shaft and hole of 90 mm size for H_8/e_9 type clearance fit. Size 90 mm falls in diameter step of 80-100. Value of tolerance unit $i(\mu) = 0.45\sqrt[3]{D} + 0.001D$. The value of tolerances for IT8 and IT9 grades are 25i and 40i respectively. Value of fundamental deviation for "e" type of shaft is $-11D^{0.41}$. Determine maximum and minimum sizes of the hole and shaft and maximum and minimum clearances. 10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
